## Type MS · Sign and magnitude numbers

### Answer 1 of 5 · Question 2 part b · 2 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(a)(i) | Award up to **two** marks for:<br><br>• 0100 (1) 1110 (1) | Accept any number of leading zeros. | 2 |
| 2(a)(ii) | **The only correct answer is B**<br><br>*A is not correct because standard ASCII uses 7 bits, 64 characters would need only 6 bits*<br>*C is not correct because standard ASCII uses 7 bits, 256 characters would need 8 bits*<br>*D is not correct because standard ASCII uses 7 bits, 512 characters would need 9 bits* | | 1 |
| 2(a)(iii) | Award up to **two** marks for a linked explanation such as:<br><br>All of the major languages/symbols/characters can be represented by Unicode (1) because it uses a minimum of 16 bits/more bits/32 bits/65536 characters (1)<br><br>Unicode can represent all/more characters/any language (1) whereas ASCII can only represent English/Latin/128 characters/doesn't have enough characters (1)<br><br>Unicode can represent all characters (1) because it uses 16 bits/2 bytes /more bits instead of 8 bits/1 byte (1) | Accept the reverse argument<br><br>Allow examples of non-latin characters | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(b) | 1010 1011<br><br>Award up to **two** marks for:<br>• MSB = 1 (1)<br>• Rest of pattern correct 010 1011 (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c) | Award up to **two** marks for:<br><br>• 0100 (1) 1010 (1) | | 2 |

## Type MS · Sign and magnitude numbers

### Answer 1 of 5 · Question 2 part b · 2 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(d)(i)</td><td>Award up to two marks for:<br><br>• 3 bits used for all patterns (1)/<br>• No pattern repeated (1)<br><br>Example:<br><br><table><tr><th>Colour</th><th>Binary pattern</th></tr><tr><td>Green</td><td>000</td></tr><tr><td>Black</td><td>001</td></tr><tr><td>White</td><td>010</td></tr><tr><td>Red</td><td>011</td></tr><tr><td>Blue</td><td>100</td></tr></table></td><td></td><td>2</td></tr>
<tr><td>2(d)(ii)</td><td>Award one mark for each of:<br><br>• 3579 x 6128 x 32 (1)<br>• ÷ 8 (1)<br>• + 732 (1)<br>• ÷ (1000 x 1000) (1)<br><br>Examples<br><br>$\frac{\frac{3579 \times 6128 \times 32}{8} + 732}{1000 \times 1000}$<br><br>$\frac{((3579 \times 6128 \times 32) \div 8) + 732}{1000 \times 1000}$<br><br>$\frac{(3579 \times 6128 \times 4) + 732}{1000 \times 1000}$</td><td>• Units are not required<br>• Equivalent expressions are accepted<br>• Calculations not explicit but expressed gain the mark<br>• Award 1 mark for correct calculated answer of 87.73 if no other marks awarded</td><td>4</td></tr>
<tr><td colspan="2"></td><td>Total for question 2</td><td>15</td></tr>
</table>

## Type MS · Sign and magnitude numbers

### Answer 1 of 5 · Question 2 part b · 2 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Sign and magnitude numbers

Answer 2 of 5 · Question 3 part e.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 4 · find part e.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 3(a)(i) | • LAN / Local area network (1)<br><br>Allow PAN/Personal Area Network | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(a)(ii) | • WAN / Wide area network (1) | | 1 |

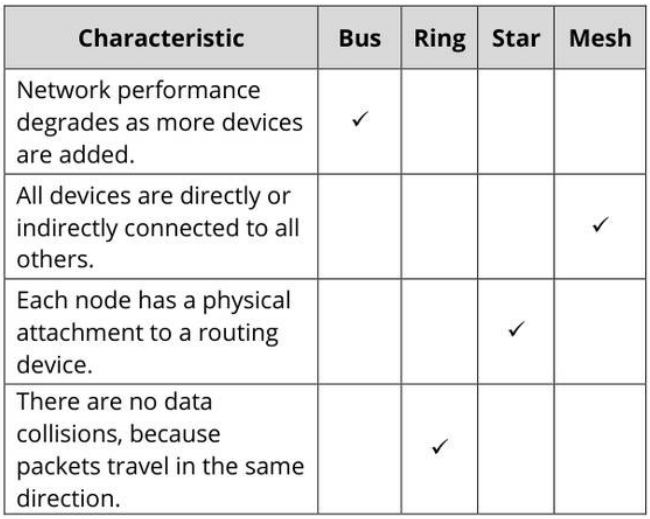
| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(b) | A: Megabits per second<br><br>**The only correct answer is A**<br><br>**B** is not correct because megapixels are associated with images rather than network data speeds<br><br>**C** is not correct because mebicycles are not associated with network data speeds<br><br>**D** is not correct because mebihertz are not associated with network data speeds | | 1 |

## Type MS · Sign and magnitude numbers

### Answer 2 of 5 · Question 3 part e.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 4 · find part e.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(c) | • IMAP (1)<br><br>AND **one** from:<br>• because changes are synced (1) with the mail server in real-time (1)<br>• because the messages always reside on the email server (1) as she has limited storage (1) on her tablet and phone<br>• it gets sent to both smartphone and tablet (1)<br><br>**Alternative answer:**<br>• POP3 (not awarded a mark)<br><br>AND **one** from:<br>• because storage limitations on server (1) and privacy issues (1)<br>• because more CPU time (1) may be used when syncing (1) (a large number of messages). | | 3 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(d) |  | | 4 |

| Characteristic | Bus | Ring | Star | Mesh |
|---|---|---|---|---|
| Network performance degrades as more devices are added. | ✓ | | | |
| All devices are directly or indirectly connected to all others. | | | | ✓ |
| Each node has a physical attachment to a routing device. | | | ✓ | |
| There are no data collisions, because packets travel in the same direction. | | ✓ | | |

## Type MS · Sign and magnitude numbers

### Answer 2 of 5 · Question 3 part e.i · 2 marks

2019 JUN · Paper 01 · extract 3 of 4 · find part e.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e)(i) | -81<br><br>One mark for negative sign<br>One mark for 81. | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e)(ii) | One mark for each nibble in correct order<br><br>0100 1011 | | 2 |

## Type MS · Sign and magnitude numbers

### Answer 2 of 5 · Question 3 part e.i · 2 marks

2019 JUN · Paper 01 · extract 4 of 4 · find part e.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e)(iii) | Any **two** from:<br><br>1. This is an overflow (1) error<br>2. It is caused by adding two 1s in the most significant bit (1)<br>3. The resulting number is too large to fit in the number of bits available / requires 9 bits / the number is larger than $2^8$ / there is 1 that needs to be carried over (in MSB) / MSB gets discarded (1)<br>4. It can cause the program to crash or produce unreliable/incorrect results (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Sign and magnitude numbers

Answer 3 of 5 · Question 2 part a.i · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(a)(i) | 1101 1110<br><br>Award one mark for:<br>• MSB = 1 (1)<br>• Rest of pattern correct (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(a)(ii) | - (1)<br>119 (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(b) | B 16 | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(i) | 128 | Allow the range 0-127 | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(ii) | C | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|



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Type MS · Sign and magnitude numbers

Answer 3 of 5 · Question 2 part a.i · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(c)(iii) | Award **two** marks for a linked explanation such as:<br><br>(Before Unicode existed) there are hundreds of different encoding systems (1) and no single encoding system could contain enough characters to represent all major languages (1)<br><br>Standard ASCII only provides 128 different patterns (1) can't represent all major languages/symbols/characters (1)<br><br>Unicode uses a minimum of 16 bits (1) so can represent at least $2^{16}$ characters (1)<br><br>Unicode has very large number of characters (1) can represent all languages/ASCII was developed (just) for English (1) | | 2 |



Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

## Type MS · Sign and magnitude numbers

### Answer 3 of 5 · Question 2 part a.i · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part a.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Sign and magnitude numbers

### Answer 4 of 5 · Question 2 part d.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part d.ii in this mark scheme extract

<table>
<tr>
<td>2(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Denary</th><th>Hexadecimal</th></tr>
<tr><td>8</td><td>8 (1)</td></tr>
<tr><td>12</td><td>C (1)</td></tr>
</table>
</td>
<td></td>
<td>2</td>
</tr>
</table>

## Type MS · Sign and magnitude numbers

### Answer 4 of 5 · Question 2 part d.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part d.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(b) | **The only correct answer is D**<br><br>***A*** *is not correct because* $6^2$ *stores 36 patterns, not 64 patterns*<br><br>***B*** *is not correct because* **6 × 2** stores 12 patterns, not 64 patterns<br><br>***C*** *is not correct because* $6^2$ **– 1** stores 35 patterns, not 64 patterns | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(i) | Award **one** mark for:<br><br>• 0001 0111 (1) | Ignore spaces | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(ii) | Award **one** mark for:<br><br>• 1111 1000 (1) | Ignore spaces | 1 |

## Type MS · Sign and magnitude numbers

### Answer 4 of 5 · Question 2 part d.ii · 2 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part d.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d)(i) | Award **one** mark for each correct nibble:<br>• 0011 1100 | Ignore spaces | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d)(ii) | Award **one** mark for each of:<br>• Negative sign (1)<br>• 22 (1)<br><br>Example:<br>• -22 | Note 22 is calculated from 16 + 4 + 2 | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d)(iii) | Award **one** mark for each correct nibble:<br>• 0001 1011 | Ignore spaces | 2 |

## Type MS · Sign and magnitude numbers

### Answer 4 of 5 · Question 2 part d.ii · 2 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part d.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(e) | Award **one** mark for each of:<br><br>1. 13 x 1000 // 13000 in the numerator (1)<br>2. 1024 in the denominator (1)<br><br>Examples:<br>$\frac{13 \times 1000}{1024}$<br><br>$\frac{13000}{1024}$<br><br>$\frac{13 \times 500}{512}$<br><br>$\frac{13 \times 250}{256}$<br><br>$\frac{13 \times 125}{128}$<br><br>$\frac{13 \times 10^3}{2^{10}}$ | Award reduced expressions, if correct<br><br>Allow all equivalent exponents<br><br>Note 1000/1024 = 1.024, so could have 13 * 1.024 | 2 |

## Type MS · Sign and magnitude numbers

### Answer 4 of 5 · Question 2 part d.ii · 2 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part d.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Sign and magnitude numbers

### Answer 5 of 5 · Question 4 part b · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(a)(i) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• It has high bandwidth/transmission speed (1)<br>• It has low latency (1)<br>• It has low energy consumption (1)<br>• It has high capacity (1)<br>• It has high security (1) | Allow 'more users can be connected' for MP4 | **2** |
| Question | Answer | Additional Guidance | Mark |
| 4(a)(ii) | Award up to **two** marks for a linked description such as:<br>• Biometric identifiers (fingerprint, iris, face) (1) matched against those stored in the app/database (1)<br>• Two-factor authentication / 2FA (1) requiring users to enter a code delivered to a separate system (1)<br>• Users can be asked for memorable information/security question (1) that only they will know the answer to (1) | System in MP2b may be on a separate device or a separate app | **2** |
| Question | Answer | Additional Guidance | Mark |
| 4(a)(iii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• To identify any bad programming practices/shortcuts/bugs (1)<br>• To highlight code vulnerabilities / enhance security (1)<br>• To make sure code is robust/will deal with exceptions (1)<br>• To make sure that good programming practices/company policies are maintained (1)<br>• To assess the efficiency of the code (1)<br>• To improve code quality (1)<br>• To share good practice/new techniques (1) | | **2** |

## Type MS · Sign and magnitude numbers

### Answer 5 of 5 · Question 4 part b · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(b)</td><td>Award one mark for each correct row:
<table>
<tr><th></th><th>8-bit binary pattern</th><th>Denary value</th></tr>
<tr><td>Lowest value</td><td>1111 1111</td><td>-127</td></tr>
<tr><td>Highest value</td><td>0111 1111</td><td>(+)127</td></tr>
</table></td><td></td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(i)</td><td>Award one mark for each correct column:
<table>
<tr><th>O</th><th>S</th><th>P</th><th>O AND S</th><th>NOT P</th><th>(O AND S) AND (NOT P)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table></td><td></td><td>3</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(ii)</td><td>Award up to two marks for a linked explanation such as:<br>• ROM is non-volatile/permanent (1) therefore it retains its contents when the power is switched off / data won't be lost when power is lost (1)<br>• The contents of ROM cannot be changed (1) which makes it suitable for storing the manufacturer's instructions (1)</td><td>Explanation can be a combination of different marking points. E.g. MP1a and MP2b<br>Do not accept MP1a and MP2a or MP1b and MP2b</td><td>2</td></tr>
<tr><td colspan="3">Total for question 4</td><td>13</td></tr>
</table>

## Type MS · Sign and magnitude numbers

### Answer 5 of 5 · Question 4 part b · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|